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course:BscIT 5thsem
Subject :java script

Getting Started with JavaScript LAB ASSIGNMENT – 03

Part 1 — Operator Precedence

Task 1.1 — Step 1: $2 ** 2 = 4$

Step 2: $3 * 4 = 12$

Step 3: $6 / 3 = 2$

Step 4: $4 + 12 = 16$

Step 5: $16 - 2 = 14$

Task 1.2 — Comparison Chaining Gotcha

```
console.log(1 < 2 < 3);
```

```
console.log(3 > 2 > 1);
```

Explanation:

```
1 < 2 < 3
```

Output: `true`

Why?

$1 < 2$ is true. In the next comparison, true is treated as 1, so it becomes $1 < 3$, which is true.

3 > 2 > 1

Output: false

Why?

First, 3 > 2 is true. true is treated as 1, so the next comparison becomes 1 > 1, which is false.

Task 1.3 — Four Different Operators

Answer: true

Reason: The expression uses arithmetic and comparison/logical operators. Arithmetic is calculated first, then comparison, and finally the logical &&, resulting in **true**.

Part 2 – Nested Ternary

Task 2.1 – Movie Ticket Pricing

Explanation:

The age is checked from the smallest condition to the larger conditions. Children below 5 get a free ticket, ages below 12 pay ₹100, adults below 60 pay ₹250, and senior citizens pay ₹150.

Task 2.2 — Shipping Cost

Cart Total	Output
₹2500	Free Shipping
₹1500	₹50
₹500	₹100

Explanation:

A cart total of ₹2000 or more gets free shipping. A total between ₹1000 and ₹1999 gets ₹50 shipping, otherwise the shipping charge is ₹100.

Task 2.3 — Discount Type

Output: VIP Discount

Explanation:

The condition checks whether the customer is a member **AND** has spent at least ₹5000. Both conditions are true, so the customer gets VIP Discount.

Part 3 — Arrays & Objects

Task 3.1 — Array

Total: 380

Average: 76

Result: Pass

Explanation:

The five scores are added together to get 380. Dividing 380 by 5 gives an average of 76. Since 76 is greater than or equal to 40, the result is Pass.

Task 3.2 — Object

4th Product: Pencil

Subtotal: ₹50

Grand Total: ₹1080

Explanation:

The fourth product was added to the existing three products. Its subtotal was calculated using price × quantity and then added to the previous grand total.

Part 4 — Type Coercion

Most surprising: NaN === NaN

Explanation:

NaN means "Not-a-Number", but it is a special JavaScript value. It is not equal to itself, so `NaN === NaN` gives false.

Task 4.2 — `typeof NaN`

Output: "number"

Explanation:

Although NaN means "Not-a-Number", JavaScript considers NaN to be a special value of the Number data type. Therefore, `typeof NaN` returns "number".

Part 5 — Smart Shopping Cart

Grand Total: ₹1030

Discount: 5%

Discount Amount: ₹51.50

After Discount: ₹978.50

GST: ₹176.13

Final Payable: ₹1154.63

Shipping: FREE

Explanation:

The grand total is ₹1030, so the 5% discount tier applies. After applying the discount, 18% GST is calculated. Shipping is free because the cart contains 3 distinct products.

Part 6 — Debugging

Snippet 1

Bugs found:

1. `const` was changed to `let` because the discount value is changed later.
2. `fnal` was a spelling mistake; it should be `discount`.

Output: `Final: 55`

Snippet 2

Bugs found:

1. "85" is a String while 85 is a Number, so === gives false.
2. Result should be result because JavaScript is case-sensitive.

Output: Pass

Why does the === bug give a wrong answer instead of an error?

"85" === 85 is a valid JavaScript comparison. The values have different data types, so JavaScript simply returns false. Therefore, it gives a wrong logical result instead of producing an error.

Snippet 3

Bugs found:

1. const was changed to let because cartTotal is changed later.
2. The : was missing in the ternary operator.

Output: Paid

Task 2.1:	script.js:12
Free	script.js:13
₹100	script.js:14
₹250	script.js:15
₹250	script.js:16
₹150	script.js:17
Task 2.2:	script.js:31
Free Shipping	script.js:32
₹50	script.js:33
₹100	script.js:34
Task 2.3: VIP Discount	script.js:51
Task 3.1:	script.js:72
Total: 340	script.js:73
Average: 68	script.js:74
Result: Pass	script.js:75
Task 3.2:	script.js:118
4th Product Subtotal: 600	script.js:119
Grand Total: 1630	script.js:120
Task 4.1:	script.js:128
53	script.js:130
2	script.js:131
NaN	script.js:132
false	script.js:133
true	script.js:134
true	script.js:135
1	script.js:136
Task 4.2:	script.js:144
number	script.js:145
Price Types:	script.js:176

1	script.js:136
Task 4.2:	script.js:144
number	script.js:145
Price Types:	script.js:176
number	script.js:177
number	script.js:178
number	script.js:179
=====	script.js:249
SHOPPING RECEIPT	script.js:250
=====	script.js:251
Notebook: ₹180	script.js:253
Pen: ₹50	script.js:254
Bag: ₹800	script.js:255
-----	script.js:257
Grand Total: ₹1030	script.js:259
Discount: 5%	script.js:260
Discount Amount: ₹51.5	script.js:261
After Discount: ₹978.5	script.js:262
GST (18%): ₹176.13	script.js:263
Final Payable: ₹1154.63	script.js:264
Shipping: FREE	script.js:266
Loyalty Points: 11.5463	script.js:271

Snippet 1 Final: 55	script.js:287
Snippet 2: Pass	script.js:299
Snippet 3: Paid	script.js:313
Live reload enabled.	(index):40